CDT504 考点速查表

Lessons 1-7 · 106 条考点 · 中英对照

先扫英文名定位，再用中文要点确认理解是否到位。

Lesson 1 • Lesson 1 Product Development Foundations

第1课 产品开发基础

1.1 Product and product development. 产品与产品开发。

产品是企业向客户销售的事物。产品开发从识别市场机会开始，到生产、销售和交付结束； 成功需要营销、设计和制造协同完成。

A product is something an enterprise sells to customers. Product development runs from recognizing a market opportunity to producing, selling and delivering a product. Success requires coordination across marketing, design and manufacturing.

1.2 Five dimensions of success. 成功的五个维度。

产品质量反映满足需求的程度，包括可靠性和稳健性；产品成本影响利润；开发时间影响响 应速度和回报；开发成本是所需投入；开发能力决定未来项目的效率与效果。

Product quality measures how well needs are met, including reliability and robustness. Product cost affects margins. Development time affects responsiveness and returns. Development cost is the investment required. Development capability determines how well future projects can be executed.

1.3 Team responsibilities. 团队职责。

营销负责细分市场、需求、目标价格和产品上市；设计负责工程方案、实体形态、人体工程学及界 面；制造负责生产系统及采购、供应链活动；财务和销售按需要参与。

Marketing identifies segments, needs, target prices and launch opportunities. Design defines engineering solutions, physical form, ergonomics and interfaces. Manufacturing designs and operates the production system, including purchasing and supply-chain activities. Finance and sales contribute as needed.

1.4 Five challenges. 五大挑战。

权衡是平衡冲突目标；动态变化要求适应技术和市场；众多细节会累积成重大影响；时间压力迫使 团队在信息不完整时决策；经济性要求创造足够价值来支持投入。

Trade-offs balance conflicting goals; dynamics require adaptation to changing technology and markets; details accumulate into major impacts; time pressure forces decisions with incomplete information; economics requires sufficient value to justify investment.

1.5 Time, cost and motivation. 周期、成本与吸引力。

不同产品的开发投入差异很大，课件以药品等说明周期可能很长；关键是合理规划。开 发工作也因创造性、满足需求、多元协作及团队精神而有吸引力。

Development effort varies greatly by product. The slides illustrate long development cycles, especially for medicines. The broader lesson is to plan realistically. The work is attractive because it involves creation, satisfying needs, diverse expertise and team spirit.

Lesson 1 • Lesson 1 Development Process

第1课 产品开发流程

1.1 Why define a process? A clear process supports quality assurance 为什么定义流程?

清晰流程支持质量保证、协调、规划、管理及改进。评审关口检查项目是否具备进入下一阶 段的条件；新证据可能要求回到前一步，而非机械地顺序推进。

, coordination, planning, management and improvement. Review gates check whether a project is ready to proceed. New evidence may require returning to an earlier activity rather than following a rigid sequence.

1.2 Six phases. 六个阶段。

规划界定机会及使命；概念开发识别需求并选择方案；系统级设计确定架构和主要子系统；详细设 计确定零件、材料和公差；测试与完善验证和改进设计；生产爬坡逐步建立稳定生产能力。

Planning defines the opportunity and mission. Concept development identifies needs and selects a solution concept. System-level design establishes architecture and major subsystems. Detail design specifies parts, materials and tolerances. Testing and refinement validate and improve the design. Production ramp-up prepares and stabilizes production.

1.3 The front-end process. 前端流程。

识别需求 → 建立目标规格 → 生成概念 → 选择概念 → 测试概念 → 确定最终规格 → 规划后 续开发。活动会重叠、迭代；经济分析、竞品标杆分析和建模贯穿前端过程。

Identify needs → establish target specifications → generate concepts → select concepts → test concepts → set final specifications → plan downstream development. Activities overlap and iterate. Economic analysis, competitive benchmarking and modeling support the entire front end.

1.4 Key outputs. 主要产出。

需求识别产出有层级、重要性标记的需求陈述；规格产出指标及数值；概念生成产出带说明的草图 或模型；选择缩小备选范围；测试提供客户证据；后续规划产出进度、人员及预算。

Needs identification produces organized need statements and importance ratings. Specifications produce metrics and values. Concept generation produces sketches or models with explanations. Selection narrows alternatives. Testing provides customer evidence. Final planning produces schedules, staffing and budgets.

1.5 The contract book. 项目契约书。

记录团队与高层之间的约定，包含使命陈述、客户需求、所选概念、规格、经济分析、进度、人 员和预算，使承诺与假设明确化。

This records agreement between the team and senior management: mission statement, customer needs, selected concept, specifications, economic analysis, schedule, staffing and budget. It makes commitments and assumptions explicit.

1.6 Models and prototypes. 模型与原型。

概念验证模型测试可行性；形态模型探索外观和人体工程学；分析或表格模型研究技术权衡；试 验原型帮助确定稳健参数。模型应服务于明确的开发问题。

Proof-of-concept models test feasibility; form models explore appearance and ergonomics; analytical or spreadsheet models explore technical trade-offs; test prototypes help establish robust parameters. A model should answer a specific development question.

Lesson 1 • Lesson 1 Process Variants and Organization

第1课 流程变体与组织

1.1 Market pull and technology push. 市场拉动与技术推动。

市场拉动从客户或市场需求出发寻找技术；技术推动从已有或新技术出发寻找有价值的 应用。通用开发流程最接近市场拉动型。

Market-pull development starts with a customer or market need and finds suitable technology. Technology-push development starts with a technology and searches for a valuable application. The generic process is closest to market pull.

1.2 Other process variants. 其他流程变体。

平台产品复用共同技术基础；工艺密集型产品要求产品和生产过程协同；定制产品按订单调整 既有设计；高风险产品优先处理重大风险；快速构建及数字产品支持快速构建—测试；复杂系统强调子系统协 调集成；产品—服务系统结合实物与服务。

Platform products reuse a shared technological base. Process-intensive products require close coordination between product and production process. Customized products adapt established designs to orders. High-risk products address major risks early. Quick-build and digital products support rapid build–test cycles. Complex systems require subsystem coordination and integration. Product-service systems combine physical goods and services.

1.3 Functional organization. 职能型组织。

人员按专业分组，便于积累深度能力及知识延续，但跨职能协调可能困难。项目型组织围绕项目 组队，跨专业协调更快，但可能削弱专业资源共享。

People are grouped by specialization. This promotes deep technical expertise and continuity, but coordination across functions can be difficult. A project organization groups people around a project, supporting fast cross-functional coordination but potentially weakening shared specialist resources.

1.4 Matrix organization. 矩阵型组织。

职能关系与项目关系并存。轻型项目经理主要协调，职能经理保留较多权力；重型项目经理拥有 更强项目权力。应根据专业深度、速度及整合需求选择结构。

Functional and project relationships coexist. A lightweight project manager mainly coordinates while functional managers retain authority. A heavyweight project manager has stronger project authority. Choose the structure according to the need for expertise, speed and integration.

1.5 Distributed teams. 分布式团队。
多地团队可提供区域市场知识、专家能力、与工厂供应商的接近性、成本优势或额外产能。通信 工具有帮助，但仍须明确职责与信息交换机制。
Multiple sites can provide regional market knowledge, specialist expertise, proximity to factories and suppliers, lower costs or additional capacity. Communication tools help, but responsibilities and information exchange still need deliberate coordination.

1.6 Practice. 练习。
绘制家庭晚餐的计划与制作流程，识别需求、约束、菜单方案、准备、测试及交付，并解释最适合的流 程类型及原因。
Map the process for planning and cooking a family dinner. Identify needs, constraints, alternative meals, preparation, testing and delivery. Explain which process variant best fits the situation and why.

Lesson 2 • Lesson 2 Opportunity Identification
第2课 机会识别

2.1 Opportunity. 机会。
机会是关于产品、需求、技术或需求与方案匹配的早期想法。它是创造价值的假设，不代表业务已被证 明能成功。
An opportunity is an early idea about a product, need, technology or possible match between a need and a solution. At this stage it is a hypothesis about creating value, not proof that a business will succeed.

2.2 Three horizons. 三个地平线。
第一地平线改进、扩展熟悉产品或降低成本；第二地平线面向相邻增长领域或核心市场下一代产 品；第三地平线探索新市场、类别或方案。分类同时考虑企业对需求和方案的熟悉程度。
Horizon 1 improves, extends or reduces the cost of familiar offerings. Horizon 2 targets adjacent growth or next-generation offerings for core markets. Horizon 3 explores new markets, categories or solutions. The framework considers familiarity with both needs and solutions.

2.3 Opportunity tournament. 机会竞赛式筛选。
生成大量原始机会，筛选后深入研究较小集合，最终选出少数优胜者。数量、平均质量与差 异程度影响发现卓越机会的可能性。早期创意需要多样性，后期筛选则集中资源。
Generate many raw opportunities, screen them, investigate a smaller set and select a few winners. Quantity, average quality and variation influence the chance of finding exceptional opportunities. Early creativity benefits from diversity; later selection concentrates resources.

2.4 Six steps. 六步法。
制定创新章程 → 生成并发现大量机会 → 筛选机会 → 深化有潜力的机会 → 选择卓越机会 → 反 思结果和过程。
Establish a charter → generate and sense many opportunities → screen opportunities → develop promising opportunities → select exceptional opportunities → reflect on results and process.

2.5 Innovation charter. 创新章程。
界定目标和边界，例如产品类别、增长目标、上市时间及渠道。它比单个项目的使命陈述更宽泛。 FroliCat希望约一年内通过现有零售渠道推出实体猫玩具。
Define objectives and boundaries, such as the product category, target growth, launch timing and channels. A charter is broader than a single project's mission statement. FroliCat sought a physical cat toy that could launch within about a year through existing retail channels.

2.6 Seven sources of ideas. 七种创意来源。
追随个人兴趣；记录令人困扰的问题清单；利用现有能力；研究客户；分析趋势；模仿并改进 ；挖掘信息来源与人际网络。
Follow a personal passion; compile a bug list of frustrations; exploit existing capabilities; study customers; consider trends; imitate but improve; and mine information sources and networks.

Lesson 2 • Lesson 2 Screening and the RWW Framework
第2课 机会筛选与RWW框架

2.1 Early screening. 早期筛选。
判断机会是否值得再投入几天或几周研究。汇集团队有依据的判断，而不是对每个初始想法都进行 昂贵细评。
Ask whether an opportunity deserves a few more days or weeks of investigation. Aggregate informed team judgments rather than conducting an expensive, highly detailed evaluation of every raw idea.

2.2 Develop promising opportunities. 深化有潜力的机会。
开展访谈、现有产品测试、初步构思、快速原型及市场估算。目标是用尽量少的时间和资 金解决最大的未知。
Use interviews, tests of existing products, initial concepts, quick prototypes and market estimates. The goal is to resolve the greatest uncertainty with the least time and money.

2.3 Real. 真实可行。
市场是否真实存在，产品是否可行？检查需求、购买意愿、市场规模、采用障碍、客户可接受风险、 技术可行性、成本及相关社会或法律约束。
Is there a genuine market and a feasible product? Examine need, willingness to buy, market size, adoption barriers, acceptable customer risk, technical feasibility, cost and relevant social or legal constraints.

2.4 Win. 能否获胜。
产品与企业能否竞争？检查持续优势、性能、价格、知识产权、时机、品牌匹配、对手可能反应、 资源、管理层投入和市场知识。
Can the product and company compete? Examine sustainable advantage, performance, price, intellectual property, timing, brand fit, competitors' likely responses, resources, management commitment and market knowledge.

2.5 Worth it. 是否值得。
预期回报是否足够，风险是否可接受？检查盈利、资金资源、技术和市场风险、战略契合及未来机 会。某一维度为“是”不能替代其他维度的证据。
Are expected returns adequate and risks acceptable? Examine profitability, available cash and resources, technical and market risk, strategic fit and future options. A positive answer to one dimension does not replace evidence for the others.

2.6 FroliCat case and reflection. FroliCat案例与反思。
团队从50个机会筛到7个概念、3个探索原型、1个正式项目。已有客户和较短上市周期 促使它聚焦第二地平线。反思应考虑学习、资源利用和机会质量，而不只看最终销量。练习提出10个机会，并 用RWW同时分析优势和风险。
The team narrowed 50 opportunities to 7 concepts, then 3 exploratory prototypes, then 1 project. It focused on Horizon 2 because it had existing customers and a short launch window. Reflect on learning, resource use and the quality of opportunities, not only eventual sales. Practice generating 10 opportunities and evaluating both strengths and risks with RWW.

Lesson 3 • Lesson 3 Product Planning and Strategy
第3课 产品规划与战略

3.1 Product planning. 产品规划。
在正式批准项目、大量投入资源前，决定推进哪些项目，以及它们如何与现有产品线、其他项目及 企业战略匹配。产出是产品计划和所选项目的使命陈述。
Before formal project approval and major resource commitment, decide which projects to pursue and how they fit the current product line, one another and business strategy. The outputs are a product plan and mission statements for selected projects.

3.2 Four project types. 四类项目。
新平台为新产品家族建立共同基础；衍生产品扩展现有平台；渐进改良修改现有功能；根本性新产 品显著偏离已有技术或市场，通常不确定性更高。
New platforms create a common base for a new product family. Derivatives extend an existing platform. Incremental improvements modify current features. Fundamentally new products depart substantially from existing technologies or markets and usually carry greater uncertainty.

3.3 Planning process. 规划流程。
识别机会 → 评价并排序项目 → 分配资源并规划时间 → 完成项目前期规划 → 反思。预算和人 员限制常使团队返回前一步调整优先级。
Identify opportunities → evaluate and prioritize projects → allocate resources and plan timing → complete pre-project planning → reflect. Budgets and staffing constraints often force the team to revisit priorities.

3.4 Four competitive strategies. 四种竞争战略。
技术领先强调研发及新技术；成本领先强调生产效率；客户导向强调变化的需求与及时响应； 模仿战略借鉴成功竞品，要求快速开发能力。
Technology leadership emphasizes research and new technology. Cost leadership emphasizes efficient production. Customer focus emphasizes changing needs and responsive offerings. An imitative strategy learns from successful competitors and requires fast development.

3.5 Segmentation. 市场细分。
将客户划分为有意义的群体，并对照本企业和竞品在各群体中的表现。寻找自身产品线空白及竞品 不足。适合某群体的产品不一定适合其他群体。
Divide customers into meaningful groups and map both company and competing products against them. Look for gaps in the company's line and weaknesses in competitors' offerings. A product that suits one segment may be unsuitable for another.

3.6 Case connections. 案例联系。
课件用移动点单、支付及积分说明整合客户体验；平台衍生说明复用；延长电池续航说明渐进改良 ； 戴森风扇说明跨入不同产品类别。
The slides use mobile ordering, payment and rewards to show an integrated customer experience; platform derivatives illustrate reuse; longer battery life illustrates incremental change; Dyson's fan illustrates a move into a different product category.

Lesson 3 • Lesson 3 Portfolio Resources and Timing

第3课 项目组合资源与时间

3.1 Technology roadmap. 技术路线图。
表示相关技术预计何时可用，以及未来哪些产品会采用，协调技术开发、子系统演进和产品推出 时间。
Map when relevant technologies are expected to be available and which future products may use them. This coordinates technology development, subsystem evolution and product introductions.

3.2 New-market evaluation. 新市场评价。
考虑年度市场规模、增长、竞争、已有市场和技术知识、产品匹配、核心能力、进入壁垒及内部 推动者。年度市场金额可近似为年销量乘以平均单价。
Consider annual market size, growth, competition, existing market and technical knowledge, product fit, core capabilities, barriers to entry and an internal champion. Annual market value can be approximated as units sold per year multiplied by average unit price.

3.3 Balanced portfolio. 均衡项目组合。
组合应服务战略，合理配置现有产品支持、渐进改良、衍生、平台和突破项目。可先向类别分 配资金，再在类别内部排序，避免把目标不同的项目直接混为一谈。
The mix should reflect strategy and include suitable levels of current-product support, incremental improvements, derivatives, platforms and breakthroughs. Allocate funding to categories first, then rank projects within each category; avoid comparing all project types as if they had identical goals.

3.4 Resource allocation. 资源分配。
按月、季度或年估算人员与设施，人时、人月是常见单位；还需考虑原型设施、测试设备和试生产 线。课件建议能力利用率约80%—90%，以保留应急及响应空间。
Estimate people and facilities by month, quarter or year. Person-hours and person-months are common measures. Consider prototype facilities, test equipment and pilot lines. The slides suggest capacity targets around 80–90% to preserve contingency and responsiveness.

3.5 Pipeline timing. 项目管线时间。
考虑上市时间、技术成熟度、市场接受准备及竞品发布。提早上市可能有利，但质量不足会损 害品牌。有预算不等于可执行，还需要合适技能和设施。
Consider launch timing, technology readiness, market readiness and competitors' launches. Earlier introduction can help, but poor quality can damage the brand. Funding alone is insufficient if the required skills or facilities are unavailable.

3.6 Planning is iterative. 规划需要迭代。
当可行性、预算、市场或优先级变化时更新组合。产品计划是持续调整的决策框架，并不是必 须执行所有初期看好项目的永久承诺。
Update the portfolio when new information changes feasibility, budgets, market conditions or priorities. A product plan is a decision framework that evolves, not a permanent promise to execute every initially attractive project.

Lesson 3 • Lesson 3 Mission Statements and Review

第3课 使命陈述与规划复核

3.1 Mission statement elements. 使命陈述要素。
包括简要产品描述、利益主张、关键业务目标、目标市场、假设和约束、利益相关者。它界定 方向，但不必规定所有技术方案。
Include a brief product description, benefit proposition, key business goals, target markets, assumptions and constraints, and stakeholders. These establish direction without specifying every technical solution.

3.2 Benefit proposition versus features. 利益主张与功能特征。
利益说明客户获得的价值；功能是可能实现该价值的产品特性。例如，减少等待是利益 ，移动点单是支持这一利益的一种功能。
A benefit explains the value received by the customer. A feature is a characteristic that may deliver that value. For example, reducing waiting is a benefit, while mobile ordering is one possible feature supporting it.

3.3 Assumptions and constraints. 假设和约束。
制造能力、服务安排、环境政策、预算及时间都可能约束项目。部分来自企业战略，而非客户明 确表达的需求。应检查约束是否必要，是否无谓限制创新。
Manufacturing capacity, service arrangements, environmental policy, budgets and timing may constrain the project. Some constraints arise from business strategy rather than expressed customer needs. Check whether each constraint is necessary or unnecessarily restricts innovation.

3.4 Pre-project commitments. 项目前期承诺。
小型跨职能核心团队明确领导、人员、预算及责任，关键成员承诺负责项目或重要部分。利益 相关者包括购买、使用、生产、销售、维护产品以及其他受影响者。
A small cross-functional core team clarifies leadership, staffing, budgets and key responsibilities. Critical members should commit to leading the project or major elements. Stakeholders include people who buy, use, produce, sell, maintain or are otherwise affected by the product.

3.5 Review questions. 复核问题。
计划是否支持战略和重要机会？资源是否充分？平台或合作能否提高资源利用率？使命要素是否一 致？团队是否接受挑战、保有足够设计自由，并能改进后续流程?
Does the plan support strategy and important opportunities? Are resources sufficient? Can platforms or partnerships improve resource use? Are the mission elements consistent? Does the team accept the challenge, retain enough freedom and have a better process for next time?

3.6 Practice. 练习。
选择一家企业，识别战略和产品线，解释其计划中可能出现哪些项目类型。分类应有依据，不应只是罗 列产品名称。
Choose a company, identify its strategy and product lines, and explain the types of projects likely to appear in its plan. Support the classification with evidence rather than merely listing product names.

Lesson 4 • Lesson 4 Understanding Customer Needs

第4课 理解客户需求

4.1 Purpose. 目的。
识别明确及潜在需求，使产品聚焦客户，以事实支持规格制定，保留记录，避免遗漏关键需求，并建立 团队共同理解。
Identify explicit and latent needs, keep the product customer-focused, support specifications with facts, preserve a record, avoid missing critical needs and build shared team understanding.

4.2 Explicit versus latent needs. 明确与潜在需求。
明确需求能够直接表达；潜在需求虽然存在，但客户在看到方案前可能尚未意识到或无法表 达。课件以手机摄像头、智能音箱和免手动尾门说明这一点。
Explicit needs can be stated directly. Latent needs exist but customers may not recognize or express them until a solution appears. The slides illustrate this with phone cameras, smart speakers and hands-free vehicle liftgates.

4.3 Five steps. 五步法。
收集客户原始数据 → 将数据解释为需求 → 建立需求层级 → 确定相对重要性 → 反思结果与过程。
Gather raw customer data → interpret the data as needs → organize needs into a hierarchy → establish relative importance → reflect on results and process.

4.4 Three collection methods. 三种收集方式。
访谈深入了解个体，常持续1—2小时；焦点小组由主持人引导约8—12位客户讨论；观察通过 被动观看或共同操作，发现真实行为及环境因素。
Interviews provide detailed individual accounts, commonly over one to two hours. Focus groups use a moderator and about 8–12 customers. Observation reveals actual behavior and context, either passively or by working alongside the user.

4.5 Choose the right participants. 选择合适对象。
购买者与使用者可能不同，应同时考虑最终用户和重要利益相关者。领先用户更早遇到需求， 可能已自创方案；极端用户有特殊用法或限制，能暴露被忽略的需求。
Purchasers and users may differ, so include end users and other important stakeholders. Lead users experience needs before the mainstream and may have invented solutions. Extreme users have unusual usage patterns or constraints that expose overlooked needs.

4.6 Nest case. Nest案例。
恒温器案例将成功与理解舒适、控制、易用、安装及节能需求联系起来。手部灵活性受限者、多分 区住宅等特殊用户，也能揭示对主流客户有价值的需求。
The thermostat example links success to understanding comfort, control, ease of use, installation and energy needs. Special users, such as people with limited dexterity or complex multi-zone homes, can reveal needs useful to mainstream customers too.

Lesson 4 • Lesson 4 Turning Evidence into Need Statements

第4课 将证据转化为需求陈述

4.1 Interview prompts. 访谈提示。
询问何时、为何使用产品，请客户演示，了解喜欢和不喜欢之处、购买标准及改进想法。采用开放 问题，避免引导客户认可预设设计。
Ask when and why the product is used, request a walkthrough, explore likes and dislikes, ask about buying criteria and invite improvement ideas. Use open prompts rather than steering the customer toward a preferred design.

4.2 Interaction quality. 互动质量。
灵活调整访谈，使用产品或视觉辅助材料，追问意外评论，观察非语言线索，在真实环境中演示， 并妥善收集和保护个人数据。
Adapt the interview, use products or visual aids, follow unexpected comments, observe nonverbal cues and encourage demonstrations in the real environment. Collect and protect personal data appropriately.

4.3 Documentation methods. 记录方式。
录音容易获取但转写耗时；笔记便于提问但应及时整理；录像记录行为，适合复核与培训；照片干 扰较小，但不能展示动态过程。
Audio is easy to capture but costly to transcribe. Notes support questioning but should be organized promptly. Video captures behavior for review and training. Photos are less intrusive but cannot show dynamic sequences.

4.4 Five writing rules. 五条写作原则。
说明产品做什么，而非如何实现；保持原始信息的具体程度；尽量采用肯定表达；写成产品属 性；在重要性评估前，避免用“必须” “应该”暗示优先级。
State what the product does, not how it does it; keep the source's level of specificity; use positive wording where possible; express a product attribute; and avoid using must or should to imply priority before importance has been assessed.

4.5 Translation example. 转化示例。
客户说“我想用手机调节恒温器”。需求可写为“无需专用设备即可远程控制恒温器”。 “可下载 的iPhone应用”是特定实现方案，不是底层需求本身。
Customer statement: I want to adjust the thermostat with my phone. Need: The thermostat can be controlled remotely without a special device. A downloadable iPhone app is a particular solution, not the underlying need.

4.6 Preserve evidence. 保留证据。
将客户原话、解释后的需求和访谈情境一并记录，让多名分析者比较理解。例如， “忘记关机”可 转化为“无人时节能”的需求，而不是对用户行为作评价。
Keep the original statement, interpreted need and interview context together. Multiple analysts can compare interpretations. For example, forgetting to turn the system off becomes a need for energy saving when nobody is home, rather than a judgment about the user.

Lesson 4 • Lesson 4 Hierarchy Importance and Human Centered

第4课 层级重要性及以人为中心的设计

4.1 Build a hierarchy. 建立层级。
每张卡片写一项需求，合并重复项，从客户角度分组，给组命名，必要时再建高层组，并复核结构。 一级需求概括方向，二级、三级增加细节。
Write one need per card, merge duplicates, group similar needs from the customer's perspective, label each group, form higher-level groups when necessary and review the organization. Primary needs summarize; secondary and tertiary needs add detail.

4.2 Thermostat hierarchy. 恒温器层级。
案例有8项一级需求、41项二级需求。一级类别为易安装、寿命长、易使用、控制精确、智能、 个性化、物有所值及可靠。星号标重要性，感叹号标潜在需求。
The example has eight primary needs and 41 secondary needs. The primary categories are easy installation, long life, ease of use, precise control, intelligence, personalization, good investment and reliability. Asterisks mark importance; an exclamation mark denotes a latent need.

4.3 Assess importance. 评估重要性。
可依据客户接触经验达成团队共识，或开展客户调查。共识更快、更省钱；调查提供直接证据， 但更费力。重点关注影响技术和成本权衡的需求，并分析平均值、差异及客户分组。
Use team consensus informed by customer contact or a customer survey. Consensus is faster and cheaper; surveys provide direct evidence but require more effort. Focus on needs that drive technical or cost trade-offs. Analyze averages, variation and different customer groups.

4.4 The example five-point scale. 案例五分量表。
1分为不希望有；2分为不重要但可以接受；3分为有则更好；4分为很希望有但非必需；5分为 不可缺少。另用勾选标出独特、令人兴奋或意外的功能。
1 means the feature is undesirable; 2 means unimportant but acceptable; 3 means nice to have; 4 means highly desirable but not essential; 5 means essential. A separate check identifies unique, exciting or unexpected features.

4.5 Reflection. 反思。
检查客户类型是否覆盖、潜在需求是否发现、哪些问题须追访、谁适合继续参与、有哪些意外发现，以 及内部人员是否充分参与。需求识别是结构化探究，不是绝对精确的科学。
Check customer coverage, latent needs, follow-up questions, useful ongoing participants, surprising findings and internal participation. Identifying needs is structured inquiry, not an exact science.

4.6 Human-centered design. 以人为中心的设计。
同理心意味着在情境中理解人，设计易用、直观的体验。儿童牙刷和小便池苍蝇图案案例 引导分析真实行为。练习找出令人困扰产品遗漏的需求，并解释遗漏原因。
Empathy means understanding people in context and designing for usable, intuitive experiences. The children's toothbrush and urinal-fly examples invite analysis of actual behavior. Practice identifying a frustrating product's missed needs and explaining why they were missed.

Lesson 5 • Lesson 5 Specifications and Metrics

第5课 规格与指标

5.1 Specification. 规格。
规格由指标及数值或允许范围组成，将主观需求转成客观目标。例如需求“易安装”，可对应指标“安 装时间”；目标数值应由客户证据、竞争和可行性支持。
A specification combines a metric with a value or allowable range. It converts a subjective need into an objective target. Need: easy to install. Metric: installation time. The target value must be justified by customer evidence, competition and feasibility.

5.2 Target versus final specifications. 目标规格与最终规格。
目标规格是在详细约束未知时提出的早期期望；最终规格在概念选择后依据技术限制、 生产成本和权衡细化，是可执行承诺，而非简单保留最激进目标。
Targets express early aspirations before detailed constraints are known. Final specifications follow concept selection and are refined using technical limits, production cost and trade-offs. They are commitments, not simply the most ambitious initial targets.

5.3 Four target-setting steps. 目标制定四步。
准备指标清单 → 收集竞争标杆信息 → 设定理想值与最低可接受值 → 反思。理想值是希望 达到的最佳结果；最低可接受值是可接受性能的边界。
Prepare the metric list → collect competitive benchmarking information → set ideal and marginally acceptable target values → reflect. An ideal value is the desired best outcome; a marginal value is the boundary of acceptable performance.

5.4 Needs–metrics matrix. 需求—指标矩阵。
行列分别列出需求与指标，以标记表示关系。一项需求可对应多个指标，一个指标也可影响 多项需求；矩阵帮助检查覆盖完整性，并支持质量功能展开QFD。
Rows list needs, columns list metrics and marks show relationships. One need may require several metrics, and one metric may influence several needs. The matrix helps check coverage and supports quality function deployment, or QFD.

5.5 Metric quality. 指标质量。
指标应完整、实际且可测，通常描述结果性的因变量，让设计者自由选择实现方式。主观需求可由 客户评议小组评价，也要考虑市场和媒体常用标准。
Metrics should be complete, practical and measurable. They should usually describe dependent outcomes, giving designers freedom to choose how to achieve them. Subjective needs may require customer panels. Include relevant criteria used in market and media evaluations.

5.6 Bicycle-fork case. 自行车避震叉案例。
需求包括减振、轻量、兼容、耐用和安全；指标包括衰减、预载、刚度、质量、配合尺寸、 安装时间、制造成本及测试寿命。重点是让每项指标对应真实需求。
Needs include vibration reduction, low mass, compatibility, durability and safe operation. Metrics include attenuation, preload, stiffness, mass, fit dimensions, assembly time, manufacturing cost and test life. The lesson is to connect each metric to a real need.

Lesson 5 • Lesson 5 Benchmarking and Target Values

第5课 标杆比较与目标数值

5.1 Two benchmarking views. 两种标杆视角。
既比较实测指标，也比较客户感知的满意度。技术数值更好未必带来更高满意度；需要核验数 据，并分析两种视角的差异。
Compare competitors using measured metrics and customer-perceived satisfaction. A technically superior result may not produce higher perceived satisfaction. Validate the data and investigate discrepancies between these two views.

5.2 Lower and upper bounds. 下限与上限。
“至少X”规定下限，例如刹车安装刚度≥325 kN/m； “至多X”规定上限，例如前叉质量≤1.4 kg。应按指标是越大越好还是越小越好确定不等号。
At least X sets a minimum, such as brake-mounting stiffness ≥325 kN/m. At most X sets a maximum, such as fork mass ≤1.4 kg. Choose the inequality according to whether more or less is desirable.

5.3 Ranges and exact values. 区间与精确值。
“X至Y之间”规定范围，例如弹簧预载480—800 N； “恰好X”规定精确目标，例如前叉偏移 量38 mm。实际制造规格通常还需考虑公差。
Between X and Y defines an interval, such as spring preload of 480–800 N. Exactly X defines a precise target, such as 38 mm rake offset. Real manufacturing specifications may also need tolerances.

5.4 Discrete options. 离散选项。
有些规格只允许指定集合，例如碗组直径1.000、1.125或1.250英寸；这不代表最小值和最大值之 间的所有数值都可接受。
Some specifications allow a defined set, such as headset diameters of 1.000, 1.125 or 1.250 inches. This differs from allowing every value in the interval between the smallest and largest options.

5.5 Interdependence and segmentation. 相互依赖与细分市场。
目标之间可能冲突或联动，应结合竞争、目标客户和使命陈述设值。不同群体可能需要 不同版本，而不是对所有人采用同一种折中产品。
Targets can conflict or interact. Use competition, the target segment and the mission statement when choosing values. Different segments may justify different versions rather than one compromise product for everyone.

5.6 Review targets. 复核目标。
检查是否有人钻指标空子、是否遗漏关键成功标准，以及目标是否具有商业意义。好指标应避免出 现数值改善却没有客户价值的情况。
Check whether people are gaming a metric, whether key success criteria are missing and whether the values are commercially meaningful. A good metric discourages artificial improvement that does not benefit customers.

Lesson 5 • Lesson 5 Final Specifications and Models

第5课 最终规格与模型

5.1 Five final-specification steps. 最终规格五步。
建立技术模型 → 建立成本模型 → 通过权衡细化规格 → 向子系统分解规格 → 反思结果与 过程。
Develop technical models → develop a cost model → refine specifications through trade-offs → flow specifications down to subsystems → reflect on results and process.

5.2 Technical models. 技术模型。
将材料、几何、油液黏度、弹簧刚度等设计决策与性能联系起来。分析模型便于快速预测；分析不 足时需实体模型。多个易管理模型可能优于一个过度复杂模型。
Link design decisions, such as materials, geometry, oil viscosity or spring stiffness, to performance. Analytical models can predict results quickly; physical models are needed when analysis is inadequate. Separate manageable models may be better than one overly complex model.

5.3 Cost models and BOM. 成本模型与BOM。
物料清单列出部件、数量和单位成本，再加入装配及其他制造费用；用高、低估计表达不确 定性，并随设计与供应商信息更新。
A bill of materials lists components, quantities and unit costs. Add assembly and other manufacturing costs; use low and high estimates to represent uncertainty. Update the model as designs and supplier information improve.

5.4 Cost calculation. 成本计算。
部件成本 = 数量×单价；总制造成本 = 各部件成本之和 + 装配成本 + 其他制造成本。前叉案例按小 时工资估算装配，并按直接成本25%计间接费用；该比例是案例假设，并非通用规则。
Component cost = quantity × unit cost. Total manufacturing cost = sum of component costs + assembly cost + other manufacturing costs. In the fork example, assembly uses an hourly rate and overhead is estimated at 25% of direct cost; that percentage is a case assumption, not a universal rule.

5.5 Trade-offs and conjoint analysis. 权衡与联合分析。
结合技术和成本模型比较可行组合，竞争图展示相对定位。联合分析用调查数据建立多属性 偏好模型，估计成熟市场中哪些组合更有吸引力。
Use technical and cost models to compare feasible combinations. Competitive maps show relative positioning. Conjoint analysis uses customer survey responses to model preferences across attributes and estimate which combinations may be attractive in established markets.

5.6 Flow-down and reflection. 向下分解与反思。
质量、成本、功耗等可加总量可分配给子系统；其他结果需要交互模型。复核整体性能、模 型不确定性、市场匹配，以及是否应为未来建立更好模型。练习为小组项目选择可测指标。
Allocate additive quantities such as mass, cost and power across subsystems. Other outcomes require models of interactions. Check total performance, model uncertainty, market fit and whether future projects need better models. Practice choosing measurable metrics for the group project.

Lesson 6 • Lesson 6 Concept Generation

第6课 概念生成

6.1 Product concept. 产品概念。
概念描述可能满足客户需求的技术、工作原理及形式，用草图、模型和简短文字表达。早期探索替 代方案，可避免后期才发现更好方案而付出高成本。
A concept describes the technology, working principles and form that may satisfy customer needs. Sketches, models and brief text communicate it. Exploring alternatives early can prevent expensive late discoveries of better solutions.

6.2 Common dysfunctions. 常见问题。
强势成员可能限制备选方案，其他产品的有用思路可能被忽略，参与不足可能削弱投入。结构化方 法拓宽搜索，并帮助经验较少的成员贡献想法。
A few dominant people may restrict alternatives; useful ideas from other products may be ignored; limited participation may weaken commitment. A structured method widens the search and helps less-experienced members contribute.

6.3 Five steps. 五步法。
澄清问题 → 外部搜索 → 内部搜索 → 系统探索 → 反思方案与过程。内外搜索可以同步开展，也 可在理解加深后重复。
Clarify the problem → search externally → search internally → explore systematically → reflect on solutions and process. External and internal search can proceed together and recur as understanding improves.

6.4 Clarify and decompose. 澄清与分解。
以使命、需求及目标规格为输入，按功能、用户操作顺序或关键需求拆分复杂问题。先聚焦创造 性方案最有价值的关键子问题。
Use the mission, needs and target specifications. Break a complex problem down by function, user-action sequence or key need. Focus initial effort on critical subproblems where creative solutions have the greatest value.

6.5 External search. 外部搜索。
访谈领先用户、咨询专家、检索专利文献、研究相关产品。先扩展再聚焦。现有方案可节省时间成 本，但采用专利方案可能需要许可。
Interview lead users, consult experts, search patents and literature, and benchmark related products. First expand the search, then focus. Existing solutions may save time and cost; using a patented solution may require licensing.

6.6 Internal search. 内部搜索。
利用个人与团队知识，暂缓批评，生成大量想法，欢迎暂时不可行的方案，并使用草图或简单模型。 结合独立构思与集体讨论，兼顾多样性及完善。
Use personal and team knowledge. Suspend criticism, generate many ideas, welcome initially infeasible ideas and use sketches or simple models. Combine independent ideation with group discussion for both diversity and refinement.

Lesson 6 • Lesson 6 Systematic Exploration

第6课 系统探索与反思

6.1 Concept fragments. 概念片段。
搜索通常得到局部解决办法，而非完整产品。关键是组织片段，将兼容要素组合为整体方案，而不 只是收集一长串想法。
Search often produces partial solutions rather than complete products. The challenge is to organize these fragments and combine compatible elements into coherent solutions, not merely to collect a long list of ideas.

6.2 Classification tree. 分类树。
将方案空间划分为不同类别，便于比较、发现遗漏分支并删去不适合类别。同一问题有不同分类角度 ，因此可以建立多棵树。
Divide the solution space into distinct categories. This supports comparison, exposes neglected branches and allows unsuitable categories to be pruned. More than one tree may be useful because the problem can be classified in different ways.

6.3 Combination table. 组合表。
列出各子问题的备选方案，再跨列探索组合。并非所有数学组合都技术合理，应依据兼容性、约束及 客户需求进行选择性组合。
List alternative solutions for each subproblem and explore combinations across columns. Not every mathematical combination is technically sensible. Use compatibility, constraints and customer needs to guide selective combination.

6.4 Exploration is iterative. 探索需要迭代。
分类和组合可能暴露知识缺口，促使进一步外部搜索、内部构思或重新界定问题。工具用于组 织创造力，而不是替代判断。
Classification and combination may reveal missing knowledge, leading to more external search, internal ideation or problem reformulation. They organize creativity rather than replace judgment.

6.5 Reflection and practice. 反思与练习。
检查关键子问题是否覆盖，是否遗漏外部知识，是否还需更多概念。为小组项目制定外部搜索计 划，写清来源、要回答的问题及所需证据。
Ask whether critical subproblems are covered, whether useful external knowledge was missed and whether additional concepts are needed. For the group project, prepare an external-search plan naming information sources, questions and evidence to collect.

Lesson 7 • Lesson 7 Concept Screening

第7课 概念筛选

7.1 Concept selection. 概念选择。
依据需求及业务标准评价方案，缩小范围，并改进或组合概念。它连接生成与测试且反复迭代，不 只是投票选一张最好看的草图。
Evaluate alternatives against needs and business criteria, narrow the set and improve or combine concepts. It is an iterative link between generation and testing, not simply a vote for the most attractive sketch.

7.2 Two stages and six steps. 两阶段与六步骤。
筛选是快速比较，评分是更详细的加权评价。两者都采用：准备矩阵 → 评价 → 排序 → 组合改进 → 选择一个或多个 → 反思。
Screening is a quick comparison; scoring is a more detailed weighted evaluation. Both use: prepare the matrix → rate concepts → rank them → combine and improve → select one or more → reflect.

7.3 Prepare the matrix. 准备矩阵。
行列分别放评价标准与概念，各概念详细程度应相当。选择熟悉产品、行业标准或代表方案作为参 考，标准同时反映客户和企业需求。
Put criteria in rows and concepts in columns. Use comparable detail for every concept. Choose a reference such as a familiar product, industry standard or representative concept. Criteria should reflect both customers and the enterprise.

7.4 Screening symbols and calculation. 筛选符号与计算。
+表示优于参考，0相同， 较差。净分 = 正号数量 负号数量。按净分排序，同时分析总分 背后的优缺点。
Use + for better than the reference, 0 for the same and − for worse. Net score = number of + ratings − number of − ratings. Rank by net score, while examining the strengths and weaknesses behind the total.

7.5 Syringe example. 注射器案例。
标准包括握持操作、易用性、设置可读性、剂量准确性、耐用性、可制造性及便携性。A—G净分 为2、 1、 2、0、1、0、0；D为参考。A和E保留，B和C淘汰，D与F组合，G修改。
Criteria include handling, usability, setting readability, dosage accuracy, durability, manufacturability and portability. A–G score 2, −1, −2, 0, 1, 0 and 0. D is the reference. A and E continue, B and C stop, D and F combine, and G is revised.

7.6 Improve before discarding. 淘汰前先考虑改进。
低分或中等分可能掩盖某项优势。案例将D和F合为DF，并减少G的笨重结构得到G+。尽量 用客观依据，并记录决策理由。
A low or average score may hide a useful strength. The case combines D and F into DF and removes bulk from G to create G+. Use objective proxies where possible and document reasons for decisions.

Lesson 7 • Lesson 7 Concept Scoring and Decisions

第7课 概念评分与决策

7.1 Weighted scoring. 加权评分。
为标准赋权，再给各方案更细的评分，例如1—5分。计算S(j) = Σ[w(i)×r(i,j)]，其中w(i)为标 准i的归一化权重，r(i,j)为概念j在该项的评分。百分比权重之和应为100%。
Assign each criterion a weight and each concept a finer rating, such as 1–5. Compute S(j) = Σ[w(i) × r(i,j)], where w(i) is criterion i's normalized weight and r(i,j) is concept j's rating. Weights expressed as percentages should sum to 100%.

7.2 Worked example for DF. DF计算示例。
权重依次为5%、15%、10%、25%、15%、20%、10%；DF评分为3、4、3、3、5、3、3。加权总分为 0.15 + 0.60 + 0.30 + 0.75 + 0.75 + 0.60 + 0.30 = 3.45。
The case uses weights 5%, 15%, 10%, 25%, 15%, 20%, 10%. DF ratings are 3, 4, 3, 3, 5, 3, 3. The weighted total is 0.15 + 0.60 + 0.30 + 0.75 + 0.75 + 0.60 + 0.30 = 3.45.

7.3 Case result. 案例结果。
A为2.75，DF为3.45，E为3.10，G+为3.05，因此DF排名第一并继续开发。A是总体参考，但各标准 可有不同参考评分；需保持评分尺度含义一致。
A scores 2.75, DF 3.45, E 3.10 and G+ 3.05, so DF ranks first and proceeds. A is the overall reference, but different criteria can have separate reference ratings. Keep the meaning of the rating scale consistent.

7.4 Reflect on the decision. 反思决策。

确认成员理解、接受结果。分歧可能揭示遗漏标准或错误评分。继续调查、再次筛选或测试，可能 比过早强求一致更合适。

Confirm that team members understand and accept the result. Disagreement may reveal missing criteria or incorrect ratings. Further investigation, another screening round or additional testing may be better than forcing premature agreement.

7.5 Benefits and limits. 益处与边界。

结构化选择提高客户导向、竞争力、可制造性、沟通、决策可追溯性和投入程度。但矩阵仍依赖 有效标准、证据、权重及评分。若已有明显占优概念，可不必细评。

Structured selection improves customer focus, competitiveness, manufacturability, communication, decision records and commitment. The matrix supports judgment; its answer still depends on valid criteria, evidence, weights and ratings. A clearly dominant concept may make detailed scoring unnecessary.

7.6 Practice. 练习。

用共同标准比较五款汽车，说明参考方案、数据和权重，计算结果，并讨论不同客户优先事项为何可能 产生不同排名。

Compare five cars using a common set of criteria. Explain the reference, data and weights, calculate results, and discuss why a different customer's priorities could produce a different ranking.